

Linear Algebra

Week 2 : Assignment 2

Detailed Solutions

Question 1

Does the set, $S = \{(1, 1), (1, 2)\}$ span $\mathbb{R}^2$?

- ✓ Yes
- No

Solution:

To determine whether the set

$$S = \{(1, 1), (1, 2)\}$$

spans $\mathbb{R}^2$, we need to check whether every vector in $\mathbb{R}^2$ can be written as a linear combination of these two vectors.

Let

$$v_1 = (1, 1), \quad v_2 = (1, 2).$$

Consider an arbitrary vector

$$(x, y) \in \mathbb{R}^2.$$

We want to find scalars a and b such that

$$a(1, 1) + b(1, 2) = (x, y).$$

Expanding,

$$(a + b, a + 2b) = (x, y).$$

Therefore,

$$a + b = x$$

and

$$a + 2b = y.$$

Subtracting the first equation from the second,

$$b = y - x.$$

Substituting into $a + b = x$,

$$a = x - b = x - (y - x) = 2x - y.$$

Thus, for every $(x, y) \in \mathbb{R}^2$,

$$(x, y) = (2x - y)(1, 1) + (y - x)(1, 2).$$

Therefore, every vector in $\mathbb{R}^2$ can be expressed as a linear combination of the vectors in S .

Another way to see this is by forming the matrix

$$A = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

Its determinant is

$$\det(A) = (1)(2) - (1)(1) = 1 \neq 0.$$

Hence, the two vectors are linearly independent. Two linearly independent vectors in $\mathbb{R}^2$ form a basis of $\mathbb{R}^2$.

Therefore,

$$\boxed{S \text{ spans } \mathbb{R}^2}.$$

Hence, the correct answer is **Yes**.

Question 2

Find the rank of the matrix

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ 4 & 5 & 9 \end{pmatrix}.$$

- ☐ 0
- ☐ 1
- ☐ 2
- ☒ 3

Solution:

We have

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ 4 & 5 & 9 \end{pmatrix}.$$

We determine the rank using elementary row operations.
Start with

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ 4 & 5 & 9 \end{pmatrix}.$$

Apply

$$R_2 \rightarrow R_2 - 2R_1$$

and

$$R_3 \rightarrow R_3 - 4R_1.$$

This gives

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 0 \\ 0 & -3 & -3 \end{pmatrix}.$$

Now apply

$$R_3 \rightarrow R_3 - 3R_2.$$

Therefore,

$$\begin{pmatrix} 1 & 2 & 3 \\ 0 & -1 & 0 \\ 0 & 0 & -3 \end{pmatrix}.$$

There are three non-zero rows.

Hence,

$$\text{rank}(M) = 3.$$

Therefore,

$$\boxed{3}$$

and option (D) is correct.

Question 3

Let M be a 2×2 matrix, with trace equal to 4 and determinant equal to 3.
Find the eigen values of M .

- $-1, -3$

- $-1, 3$
- $1, -3$
- ✓ $1, 3$

Solution:

Let the eigenvalues of M be

$$\lambda_1, \lambda_2.$$

For any square matrix, the sum of the eigenvalues is equal to the trace of the matrix. Therefore,

$$\lambda_1 + \lambda_2 = \text{tr}(M) = 4.$$

Also, the product of the eigenvalues is equal to the determinant:

$$\lambda_1 \lambda_2 = \det(M) = 3.$$

Thus, we need two numbers whose sum is 4 and whose product is 3. The numbers are

$$1 \quad \text{and} \quad 3.$$

Indeed,

$$1 + 3 = 4$$

and

$$1 \times 3 = 3.$$

Alternatively, the characteristic polynomial is

$$\lambda^2 - (\text{tr } M)\lambda + \det(M) = 0.$$

Therefore,

$$\lambda^2 - 4\lambda + 3 = 0.$$

Factoring,

$$(\lambda - 1)(\lambda - 3) = 0.$$

Hence,

$$\lambda = 1, \quad 3.$$

Therefore,

$$\boxed{\lambda_1 = 1, \quad \lambda_2 = 3}.$$

Hence, option (D) is correct.

Question 4

Which among the following can be the eigen vector for the matrix

$$A = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix}?$$

- $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
 - ✓ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
 - $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$
 - $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$
-

Solution:

A non-zero vector v is an eigenvector of a matrix A if there exists a scalar λ such that

$$Av = \lambda v.$$

We test the given vectors.

Consider

$$v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Then

$$Av = \begin{pmatrix} 1 & 4 \\ 2 & 3 \end{pmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Multiplying,

$$Av = \begin{bmatrix} 1(1) + 4(1) \\ 2(1) + 3(1) \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}.$$

Therefore,

$$Av = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Thus,

$$\lambda = 5.$$

Since the resulting vector is a scalar multiple of the original vector, $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ is an eigenvector.

Therefore,

$$\boxed{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}$$

is the correct answer.

Hence, option (B) is correct.

Question 5

Let

$$A = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}.$$

Suppose the eigen values corresponding to AA^T are a, b, c , then find the value of $ab + bc + ca$.

- ☐ 9
- ☒ 0
- ☐ 81
- ☐ 18

Solution:

We are given

$$A = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix}.$$

First calculate AA^T .

Since

$$A^T = \begin{bmatrix} -1 & 2 & 2 \end{bmatrix},$$

we get

$$AA^T = \begin{bmatrix} -1 \\ 2 \\ 2 \end{bmatrix} \begin{bmatrix} -1 & 2 & 2 \end{bmatrix}.$$

Therefore,

$$AA^T = \begin{pmatrix} 1 & -2 & -2 \\ -2 & 4 & 4 \\ -2 & 4 & 4 \end{pmatrix}.$$

The matrix AA^T is an outer product of a non-zero vector with its transpose. Therefore, it has rank 1.

Hence, two of its three eigenvalues are zero.

The only non-zero eigenvalue equals the trace of AA^T (equivalently, $A^T A$):

$$A^T A = (-1)^2 + 2^2 + 2^2 = 1 + 4 + 4 = 9.$$

Therefore, the eigenvalues are

$$a, b, c = 9, 0, 0$$

in some order.

Now calculate

$$ab + bc + ca.$$

Since two eigenvalues are zero, every product contains a zero:

$$ab + bc + ca = (9)(0) + (0)(0) + (0)(9).$$

Thus,

$$ab + bc + ca = 0.$$

Therefore,

$$\boxed{0}$$

and option (B) is correct.

Question 6

Which among the following is true for an $n \times n$ matrix A ?

- If λ is an eigen value of A , then $1/\lambda$ is an eigen value of A^{-1} .
- $\det(A^{-1}) = (\det(A))^{-1}$

- Let B be an $n \times n$ matrix. Then $\det(AB) = \det(A) \det(B)$
- ✓ All of the above

Solution:

We verify each statement.

Statement 1:

Suppose λ is an eigenvalue of A . Then there exists a non-zero vector v such that

$$Av = \lambda v.$$

Assuming A is invertible, multiply both sides by A^{-1} :

$$A^{-1}Av = A^{-1}\lambda v.$$

Therefore,

$$v = \lambda A^{-1}v.$$

Dividing by λ ,

$$A^{-1}v = \frac{1}{\lambda}v.$$

Thus,

$$\boxed{\frac{1}{\lambda}}$$

is an eigenvalue of A^{-1} .

Hence, statement 1 is true.

Statement 2:

For an invertible matrix,

$$AA^{-1} = I.$$

Taking determinants,

$$\det(A) \det(A^{-1}) = \det(I).$$

Since

$$\det(I) = 1,$$

we get

$$\det(A^{-1}) = \frac{1}{\det(A)} = (\det(A))^{-1}.$$

Hence, statement 2 is true.

Statement 3:

The determinant of a product satisfies

$$\det(AB) = \det(A) \det(B).$$

Hence, statement 3 is also true.

Therefore, all three statements are correct.

Thus,

All of the above

and option (D) is correct.

Question 7

Check if the following set of vectors are linearly independent.

$$v_1 = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \quad v_3 = \begin{bmatrix} 3 \\ -3 \\ 8 \end{bmatrix}.$$

○ Yes

✓ No

Solution:

Three vectors are linearly independent if the only solution to

$$c_1 v_1 + c_2 v_2 + c_3 v_3 = 0$$

is

$$c_1 = c_2 = c_3 = 0.$$

We can check linear independence by forming a matrix using the vectors as columns:

$$A = \begin{pmatrix} 2 & 1 & 3 \\ -1 & 1 & -3 \\ 3 & -2 & 8 \end{pmatrix}.$$

The vectors are linearly independent if

$$\det(A) \neq 0.$$

Calculate the determinant:

$$\det(A) = 2 \begin{vmatrix} 1 & -3 \\ -2 & 8 \end{vmatrix} - 1 \begin{vmatrix} -1 & -3 \\ 3 & 8 \end{vmatrix} + 3 \begin{vmatrix} -1 & 1 \\ 3 & -2 \end{vmatrix}.$$

Now,

$$\begin{vmatrix} 1 & -3 \\ -2 & 8 \end{vmatrix} = (1)(8) - (-3)(-2) = 8 - 6 = 2.$$

Also,

$$\begin{vmatrix} -1 & -3 \\ 3 & 8 \end{vmatrix} = (-1)(8) - (-3)(3) = -8 + 9 = 1.$$

And,

$$\begin{vmatrix} -1 & 1 \\ 3 & -2 \end{vmatrix} = (-1)(-2) - (1)(3) = 2 - 3 = -1.$$

Therefore,

$$\det(A) = 2(2) - 1(1) + 3(-1).$$

Hence,

$$\det(A) = 4 - 1 - 3 = 0.$$

Since the determinant is zero, the vectors are linearly dependent.

Therefore,

No

and option (B) is correct.

Question 8

Consider the following system of linear equations:

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &= 1, \\ 2x_1 + 5x_2 + 7x_3 &= 2, \\ 2x_1 + 4x_2 + 6x_3 &= 2. \end{aligned}$$

Choose the correct option.

- ✓ It has infinite number of solutions.
 - It has a unique solution.
 - It has no solution.
-

Solution:

The system can be written in matrix form as

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 2 & 4 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}.$$

Notice that the third equation is exactly twice the first equation:

$$2(x_1 + 2x_2 + 3x_3) = 2.$$

Thus, the third equation does not provide any new independent information.

The first and second equations are independent, since the second row is not a scalar multiple of the first row.

Therefore, there are only two independent equations for three unknowns.

Hence,

$$\text{rank}(A) = 2$$

and

$$\text{rank}([A|B]) = 2.$$

Since

$$\text{rank}(A) = \text{rank}([A|B]) = 2$$

and the number of unknowns is

$$n = 3,$$

we have

$$\text{rank}(A) < n.$$

Therefore, the system has infinitely many solutions.

Thus,

It has infinite number of solutions.

Hence, option (A) is correct.

Question 9

Which of the following set of vectors are orthogonal? *(Select all that apply.)*

$$\begin{array}{ll}
\boxtimes & \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \quad \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \\
\Box & \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}, \quad \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix} \\
\Box & \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}, \quad \begin{bmatrix} -2 \\ 5 \\ 2 \end{bmatrix} \\
\boxtimes & \begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix}, \quad \begin{bmatrix} -2 \\ 4 \\ 6 \end{bmatrix}
\end{array}$$

Solution:

Two vectors are orthogonal if and only if their dot product is zero.
Therefore, we calculate the dot product for each pair.

Pair 1:

Let

$$u = \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \quad v = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Then

$$u \cdot v = (1)(2) + (-1)(0) + (-2)(1).$$

Therefore,

$$u \cdot v = 2 + 0 - 2 = 0.$$

Hence, this pair of vectors is orthogonal.

Pair 2:

$$\begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 4 \\ 2 \end{bmatrix}$$

gives

$$(1)(-1) + (-2)(4) + (4)(2) = -1 - 8 + 8 = -1.$$

Since

$$-1 \neq 0,$$

this pair of vectors is not orthogonal.

Pair 3:

$$\begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 5 \\ 2 \end{bmatrix}$$

gives

$$(1)(-2) + (-2)(5) + (4)(2) = -2 - 10 + 8 = -4.$$

Since

$$-4 \neq 0,$$

this pair of vectors is not orthogonal.

Pair 4:

$$\begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} -2 \\ 4 \\ 6 \end{bmatrix}$$

gives

$$(5)(-2) + (-2)(4) + (3)(6).$$

Therefore,

$$-10 - 8 + 18 = 0.$$

Hence, this pair of vectors is also orthogonal.

Therefore, the correct choices are pairs 1 and 4:

$$\boxed{\begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 5 \\ -2 \\ 3 \end{bmatrix}, \begin{bmatrix} -2 \\ 4 \\ 6 \end{bmatrix}}$$

Both pairs have dot product equal to zero.

Question 10

Suppose the eigen values of matrix A are $-3, -4, -4, -3$, then the determinant of the matrix $(A^{-1})^T$ is

- 0.06
- -14
- 0

✓ 0.0069

Solution:

The eigenvalues of A are

$$-3, \quad -4, \quad -4, \quad -3.$$

The determinant of a matrix is equal to the product of its eigenvalues.

Therefore,

$$\det(A) = (-3)(-4)(-4)(-3).$$

First,

$$(-3)(-4) = 12$$

and

$$(-4)(-3) = 12.$$

Hence,

$$\det(A) = 12 \times 12 = 144.$$

For an invertible matrix,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Therefore,

$$\det(A^{-1}) = \frac{1}{144}.$$

We are asked to find

$$\det((A^{-1})^T).$$

The determinant of a transpose is the same as the determinant of the original matrix:

$$\det(B^T) = \det(B).$$

Therefore,

$$\det((A^{-1})^T) = \det(A^{-1}).$$

Hence,

$$\det((A^{-1})^T) = \frac{1}{144}.$$

Numerically,

$$\frac{1}{144} = 0.006944 \dots$$

which is approximately

$$0.0069.$$

Therefore,

$$\boxed{0.0069}$$

and option (D) is correct.
